File

Name: K05-EV-stage2

Changed by: AAA on: 28.10.2026 at: 22:07:13

1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		z3(z3z4)	
Gear 2:		z4(z3z4)	
		------- Gear 1 ----------	Gear 2 -------
Power (kW)	[P]	234.572	
Speed (1/min)	[n]	2983.6	813.7
Number of load cycles (in mio.)	[NL]	1700.656	463.815
Torque (Nm)	[T]	750.8	2752.8
Application factor	[KA]	1.00	
Required service life (h)	[H]	9500.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Left flank	
Gear 1 direction of rotation:		Counterclockwise	

2 Tooth geometry

Geometry calculation according to ISO 21771:2007 (withdrawn)

		------- Gear 1 ----------	Gear 2 -------
Center distance (mm)	[a]	150.200	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	2.5000	
Transverse module (mm)	[mt]	2.6441	
Normal diametral pitch (1/in)	[Pnd]	10.16000	
Transverse diametral pitch (1/in)	[Ptd]	9.60647	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	19.0000	
Number of teeth	[z]	24	88
Facewidth (mm)	[b]	40.00	38.00
Hand of gear		right	left
Accuracy grade	[Q-ISO 1328:1995]	6	6
Inner diameter (mm)	[di]	28.00	210.14
Shaft diameter (mm)	[ds]	28.00	210.14
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1 ----------	Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

------- Gear 1 ---------- Gear 2 -------

Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1 ----------	Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	ISO-VG 46	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	46.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	6.80
Specific density at 15°C (kg/dm³)	[ρ]	0.878
Oil temperature (°C)	[TS]	70.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.666
Root radius factor	[ρfP*]	0.203
	[ρfPmax*]	0.256
Addendum coefficient	[haP*]	1.176
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	32.849
(mm)	[a0.e/i]	32.753 /32.698
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.710 / 0.719

Reference profile Gear 2

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.371
Root radius factor	[ρfP*]	0.356
	[ρfPmax*]	0.409
Addendum coefficient	[haP*]	1.456
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000

	not topping	
Manufacturing center distance (mm)	[a0]	117.450
(mm)	[a0.e/i]	117.319 /117.251
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.911 / 0.913

4.1.1 Information on final machining

		------- Gear 1	---------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.666	1.371
Tooth root radius reference profile	[ρfP*]	0.203	0.356
Addendum reference profile	[haP*]	1.176	1.456
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	for high load capacity gearboxes		
Tip relief (µm)	[Ca L/R]	0.0 / 0.0	0.0 / 0.0

4.2 Basic data

Overall transmission ratio	[itot]	-3.667
Gear ratio	[u]	3.667
Transverse module (mm)	[mt]	2.644
Transverse pressure angle (°)	[αt]	21.054
Working pressure angle (°)	[αwt]	23.075
(°)	[αwt.e/i]	23.093 /23.058
Working pressure angle at normal section (°)	[αwn]	21.910
Base helix angle (°)	[βb]	17.814
Helix angle at operating pitch circle (°)	[βw]	19.254
Reference center distance (mm)	[ad]	148.067
Sum of profile shift coefficients	[Σxi]	0.8929

		------- Gear 1	---------- Gear 2 -------
Profile shift coefficient	[x]	0.4483	0.4446
Generating profile shift coefficient	[xE.e/i]	0.4098/ 0.3879	0.3924/ 0.3649
Virtual gear no. of teeth	[zn]	28.004	102.681
Involute length (mm)	[l_dFa-l_dFf]	6.573	6.945

4.3 Diameters and their allowances

		------- Gear 1	---------- Gear 2 -------
Reference diameter (mm)	[d]	63.457	232.677
Base diameter (mm)	[db]	59.221	217.144
Tip alteration (mm)	[k*mn]	0.000	0.000
Tip diameter (mm)	[da]	71.579	242.182
(mm)	[da.e/i]	71.579 /71.549	242.182 /242.136
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.030	0.000 / -0.046
Tip form diameter (mm)	[dFa]	71.579	242.182
(mm)	[dFa.e/i]	71.579 /71.549	242.182 /242.136
Active tip diameter (mm)	[dNa]	71.579	242.182
(mm)	[dNa.e/i]	71.579 /71.549	242.182 /242.136
V-Circle diameter (mm)	[dv]	65.699	234.900
(mm)	[dv.e/i]	65.506 /65.397	234.639 /234.501

Operating pitch diameter (mm)	[dw]	64.371	236.029
(mm)	[dw.e/i]	64.380 /64.363	236.060 /235.997
Root diameter (mm)	[df]	57.370	228.042
(mm)	[df.e/i]	57.178 /57.068	227.781 /227.644
Active root diameter (mm)	[dNf]	60.144	230.571
(mm)	[dNf.e/i]	60.181 /60.127	230.623 /230.537
Root form diameter (mm)	[dFf]	59.721	229.390
(mm)	[dFf.e/i]	59.654 /59.618	229.157 /229.035

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.389	0.424
Effective tip clearance (mm)	[c.e/i]	0.624 / 0.500	0.618 / 0.500
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.281 / 0.236	0.794 / 0.690
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	4.061	4.753
(mm)	[ha.e/i]	4.061 / 4.046	4.753 / 4.730
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	3.044	2.317
(mm)	[hf.e/i]	3.140 / 3.195	2.448 / 2.516
Tooth height (mm)	[h]	7.105	7.070

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	38.898 /38.846	28.297 /28.269
Roll angle to dNf (°)	[ξNf.e/i]	10.356 /10.058	20.499 /20.431
Roll angle at dFf (°)	[ξFf.e/i]	6.943 / 6.647	19.320 /19.219

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8971	1.8944
Normal tooth thickness at tip circle (mm)	[san]	0.938	0.883
(mm)	[san.e/i]	0.879 / 0.815	0.806 / 0.733
Normal tooth thickness at tip form circle (mm)	[sFan]	0.938	0.883
(mm)	[sFan.e/i]	0.879 / 0.815	0.806 / 0.733
Normal space width at root circle (mm)	[efn]	0.000	1.525
(mm)	[efn.e/i]	0.000 / 0.000	1.538 / 1.545
Pitch on reference circle (mm)	[pt]	8.307	8.307
Base pitch (mm)	[pbt]	7.752	7.752
Transverse pitch on contact-path (mm)	[pet]	7.752	7.752
Lead height (mm)	[pz]	578.974	2122.905
Axial pitch (mm)	[px]	24.124	24.124

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	2.977	2.929
Specific sliding at the tip	[ζa]	0.474	0.641
Specific sliding at the root	[ζf]	-1.786	-0.901

Mean specific sliding	[ζm]	0.557	
Sliding factor on tip	[Kga]	0.296	0.291
Sliding factor on root	[Kgf]	-0.291	-0.296

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	75.575
Transverse contact ratio	[εα]	1.916
	[εα.e/m/i]	1.923 / 1.911/ 1.899
Overlap ratio	[εβ]	1.575
Total contact ratio	[εγ]	3.491
	[εγ.e/m/i]	3.498 / 3.486/ 3.475
Length of path of contact (mm)	[gα]	14.854
(mm)	[gα.e/m/i]	14.905 /14.814/14.724
Approach path of contact (mm)	[gf]	7.366
(mm)	[gf.e/m/i]	7.406 / 7.340/ 7.274
Recess path of contact (mm)	[ga]	7.488
(mm)	[ga.e/m/i]	7.499 / 7.474/ 7.450

		Gear 1	Gear 2
Addendum contact ratio	[ε]	0.966	0.950
	[ε.e/i]	0.967 / 0.961	0.955 / 0.938
Length T1-A and T2-A (mm)	[T1A,T2A]	5.249	53.621
(mm)	[.e/i]	5.198 / 5.352	53.621 / 53.569
Length T1-B and T2-B (mm)	[T1B,T2B]	12.351	46.519
(mm)	[.e/i]	12.351 / 12.324	46.468 / 46.597
Length T1-C and T2-C (mm)	[T1C,T2C]	12.615	46.255
(mm)	[.e/i]	12.604 / 12.626	46.215 / 46.295
Length T1-D and T2-D (mm)	[T1D,T2D]	13.001	45.869
(mm)	[.e/i]	12.950 / 13.104	45.869 / 45.817
Length T1-E and T2-E (mm)	[T1E,T2E]	20.103	38.767
(mm)	[.e/i]	20.103 / 20.076	38.716 / 38.845
Length T1-T2 (mm)	[T1T2]	58.870	
(mm)	[.e/i]	58.819 / 58.921	
Diameter of single contact point B (mm)	[d-B]	64.166	236.236
(mm)	[d-B.e/i]	64.166 / 64.145	236.196 /236.298
Diameter of single contact point D (mm)	[d-D]	64.678	235.727
(mm)	[d-D.e/i]	64.637 / 64.761	235.727 /235.687

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	23662.21	
Axial force (N)	[Fa]	8147.55	8147.55
Radial force (N)	[Fr]	9108.59	9108.59
Normal force (N)	[Fnorm]	26631.73	26631.73
Nominal circumferential force per mm (N/mm)	[w]	622.69	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	23326.16	
Axial force (N)	[Faw]	8147.55	8147.55

Radial force (N)	[Frw]	9937.64
Circumferential speed reference circle (m/s)	[v]	9.91
Circumferential speed operating pitch circle (m/s)	[v(dw)]	10.06

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.6
Running-in value (µm)	[yf]	0.7
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	0.841
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	12.693
Theoretical single stiffness (N/mm/µm)	[c'th]	19.960
Contact stiffness (N/mm/µm)	[cγα]	21.414
Contact stiffness (N/mm/µm)	[cγβ]	18.202
Reduced mass (kg/mm)	[mRed]	0.01215
Resonance speed (min-1)	[nE1]	16705
Resonance ratio (-)	[N]	0.179

Subcritical range

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.4

5.4 K factors

Dynamic factor	[Kv]	1.024

Factor $K_{H\beta}$, own input

Face load factors

- Flank	[KHβ]	1.055
- Tooth root	[KFβ]	1.045
- Scuffing	[KBβ]	1.055

Transverse load factors

- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000

Application factor	[KA]	1.000
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.3879	0.3649
Tooth form factor	[YF]	0.785	0.837
Stress correction factor	[YS]	2.305	2.330
Load application angle (°)	[αFen]	17.743	20.961
Load distribution influence factor	[fε]	0.688	
Load application diameter, virtual spur gear (mm)	[d_{en}]	70.632	259.681
Load application diameter (mm)	[d_{en}]	64.079	235.655
Bending moment arm (mm)	[hFe]	2.499	2.909
(-)	[hFe*]	0.999	1.163
Tooth root thickness at critical cross section (mm)	[sFn]	5.768	5.968
(-)	[sFn*]	2.307	2.387
Root fillet radius at critical cross section (mm)	[ρF]	1.118	1.014

	(-)	[ρF*]	0.447	0.406
Tangent contact point, virtual spur gear (mm)		[x,y]	2.884 /28.762	
Tangent contact point, virtual spur gear (mm)		[x,y]		2.984 /114.236
Diameter at critical cross section (mm)		[d_{sFn}]	57.812	228.551
Tangent at critical cross section (°)		[α_{sFn}]	30.000	30.000
Notch parameter		[q_s]	2.579	2.943
Helix angle factor		[Yβ]	0.996	
Deep tooth factor		[YDT]	1.000	
Gear rim factor		[YB]	1.00	1.00
Effective facewidth (mm)		[beff]	40.00	38.00
Nominal stress at tooth root (N/mm²)		[σF0]	426.51	483.97
Tooth root stress (N/mm²)		[σF]	456.03	517.47
Permissible bending stress at root of Test-gear				
Notch sensitivity factor		[YδrelT]	1.001	1.004
Surface factor		[YRrelT]	0.957	0.957
Size factor, tooth root		[YX]	1.000	1.000
Life factor		[YNT]	0.881	0.904
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$			0.843	0.868
Alternating bending factor, mean stress influence coefficient				
		[YM]	1.000	1.000
Stress correction factor		[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)		[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)		[σFP]	511.97	527.20
Limit strength tooth root (N/mm²)		[σFG]	716.75	738.07

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	1.57	1.43
Transmittable power (kW)	[kWRating]	263.34	238.98

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.265	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.722	
Helix angle factor	[Zβ]	1.028	
Effective facewidth (mm)	[beff]	38.00	
Nominal contact stress (N/mm²)	[σH0]	1128.95	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1172.99	
Coefficient [fZCa] 1.07 (Helical gear sets with suitable flank modifications acc. to manufacturers experience)			
Single tooth contact factor	[ZB,ZD]	1.03	1.03
Contact stress (N/mm²)	[σHB, σHD]	1213.35	1213.35
Lubricant coefficient for N_L	[ZL]	0.931	0.931
Speed factor at N_L	[ZV]	1.000	1.000
Roughness factor for N_L	[ZR]	0.963	0.963
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.897	0.934
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.805	0.837
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1206.83	1255.90
Pitting stress limit (N/mm²)	[σHG]	1206.83	1255.90

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.03	1.07
Safety against pressure, σHG/σHBD Single contact	[SHBD]	0.99	1.04
Safety regarding transmittable torque	[SHBD²]	0.99	1.07
Transmittable power (kW)	[kWRating]	232.06	251.31

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Calculation has not been carried out, lubricant: Load stage micropitting test not known

9 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	0.00	0.00
Optimal tip relief (µm)	[Ceff]	29.08	
C_a taken as optimal in the calculation. 0=no, 1=yes		1	1

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.300	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	12
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	38.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]	672.224	
$K_{B\gamma}$ = 1.300, $w_{Bt} \cdot K_{B\gamma}$ = 873.888			
Angle factor	[Xαβ]	1.011	
ε_1 =0.966, ε_2 =0.950			

9.1 Flash temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	0.881		
Tooth mass temperature (°C)	[θMi]	82.88		
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]	27.39		
Scuffing temperature (°C)	[θS]	365.60		
Γ coordinates (point of highest temperature)	[Γ]	-0.346		
[Γ.A]= -0.584, [Γ.E]= 0.594				
Maximum flash temperature (Δ°C)	[θflamax]	46.86		
Maximum contact temperature (°C)	[θBmax]	129.73		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Approach factor	[XJ]	1.000		
Load sharing factor	[XΓ]	0.580		
Dynamic viscosity (mPa*s)	[ηM]	12.46	(70.0	°C)
Coefficient of friction	[μ_m]	0.063		
Required safety	[SBmin]	2.000		
Margin of safety for scuffing, flash temperature	[SB]	4.949		

9.2 Integral temperature-criteria

		---------- Pair ----------

Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	80.73	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	365.79	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.225	
Dynamic viscosity (mPa*s)	[ηOil]	12.46	(70.0 °C)
Mean coefficient of friction	[μ_m]	0.069	
Geometry factor	[XBE]	0.257	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.894	
Mean flash temperature (Δ°C)	[θflaint]	15.33	
Integral tooth flank temperature (°C)	[θint]	103.72	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	3.527	
Safety factor for transmitted moment (int.-T.)	[SSL]	8.773	

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.070 / -0.110	-0.095 / -0.145

10.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	11.000
Base tangent length (no backlash) (mm)	[Wk]	27.584	81.868
Base tangent length with allowance (mm)	[Wk.e/i]	27.518 /27.480	81.779 /81.732
(mm)	[ΔWk.e/i]	-0.066 / -0.103	-0.089 / -0.136
Diameter of measuring circle (mm)	[dMWk.m]	64.750	230.673

10.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	4.663	4.302
Effective diameter of ball/pin (mm)	[DMeff]	5.000	5.000
Radial single-ball measurement, no backlash (mm)	[MrK]	36.724	121.651
Radial single-ball measurement (mm)	[MrK.e/i]	36.655 /36.616	121.537 /121.477
Diameter of measuring circle (mm)	[dMMr.m]	66.032	236.105
Diametral measurement over two balls, no backlash (mm)	[MdK]	73.449	243.302
Diametral measurement over two balls (mm)	[MdK.e/i]	73.311 /73.231	243.074 /242.954
Diametral measurement over pins, no backlash (mm)	[MdR]	73.449	243.302
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	73.311 /73.231	243.074 /242.954
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	73.311 /73.231	243.074 /242.954

10.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	71.564	242.159
Reference chordal height from da.m (mm)	[hac]	4.133	4.763
Tooth thickness at height hac, chord (mm)	[sc]	4.739	4.736
(mm)	[sc.e/i]	4.671 / 4.632	4.641 / 4.592
Tooth thickness on reference circle, arc (mm)	[sn]	4.743	4.736

(mm)	[sn.e/i]	4.673 / 4.633	4.641 / 4.591

10.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.017 / -0.017
Radial backlash (mm)	[jrw.e/i]	0.343 / 0.189
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.291 / 0.160
Normal backlash (mm)	[jnw.e/i]	0.253 / 0.141
Center distance allowances (mm)	[Aa.e/i]	0.020 / -0.020
Backlash free center distance (mm)	[aControl.e/i]	149.991 / 149.877
Backlash free center distance, allowances (mm)	[jta]	-0.209 / -0.323

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	59.877	230.019
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	0.111	0.431
Tip clearance (mm)	[c0.i(aControl)]	0.197	0.197

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.1411 / 0.0777	(0°8'28" / 0°4'40")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.5174 / 0.2848	(0°31'2" / 0°17'5")

11 Gear tolerances

		Gear 1	Gear 2
According to ISO 1328-1:1995, ISO 1328-2:1997			
Accuracy grade	[Q]	6	6
Single pitch tolerance (µm)	[fpT]	±8.50	±9.00
Base pitch tolerance (µm)	[fpbT]	±7.90	±8.40
Sector pitch tolerance (µm)	[Fpz/8T]	±13.00	±21.00
Cumulative pitch tolerance, total (µm)	[FpT]	27.00	35.00
Profile form tolerance (µm)	[ffαT]	8.50	9.50
Profile slope tolerance (µm)	[fHαT]	±7.00	±8.00
Profile tolerance, total (µm)	[FαT]	11.00	13.00
Helix form tolerance (µm)	[ffβT]	8.50	9.00
Helix slope tolerance (µm)	[fHβT]	±8.50	±9.00
Helix tolerance, total (µm)	[FβT]	12.00	13.00
Runout tolerance (µm)	[FrT]	21.00	28.00
Single flank composite tolerance, total (µm)	[FisT]	38.00	47.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	11.00	12.00
Radial composite tolerance, total (µm)	[Fi"]	31.00	37.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	9.50	9.50

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R39	R40
Radial composite tolerance, total (µm)	[FidT]	29.00	41.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	13.00	17.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99		
Total helix deviation for calculation (µm)	[Fβ]	13.00
Distance between bearings for calculation (mm)	[LG]	80.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 13.68
Maximum value for axis inclination error (µm)	[fΣδ]	± 27.37

12 Modifications and determination of the tooth form

12.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

13 Manufacturing

13.1 Power skiving

13.1.1 Power skiving, Gear 1

Power Skiving: Must be checked more closely

Machine, own input			
Minimum tool diameter	[$d_{tool,min}$]	0.0000	
Maximum tool diameter	[$d_{tool,max}$]	245.0000	
Skiving tool number of teeth	[z0]	49	
Skiving tool helix angle (°)	[β0]	39.0000	left
Skiving tool tip diameter (mm)	[da0]	157.6280	
Estimated power skiving tool shank diameter (mm)	[d_s]	135.0000	
Axial crossing angle for manufacturing (°)	[Σ0]	20.0000	
Calculation for tooth thickness/diameter tolerance field		1 / 1	*
(0: min, 1: mean, 2: max, 3: nominal)			

- Collision with tool front face: Needs to be checked in more detail

Groove diameter (mm)	[D_l]	60.0000	
Groove width (mm)	[L_l]	10.0000	
Minimum groove width (mm)	[$L_{l,min}$]	9.3641	

13.2 Honing

13.2.1 Honing, Gear 1

Honing: Not possible with the input data

Cause:

- The clamping diameter is 0.

Clamping: in the bore

Clamping diameter (mm)	[d_{clam}]	0.0000	
Required surface roughness (µm)	[R_a]	0.0000	
Required flank line crowning (µm)	[$C_β$]	0.0000	

Collision with tool front face: Apparently none

Groove diameter (mm)	[D_l]	60.0000	
Groove width (mm)	[L_l]	10.0000	
External diameter of honing wheel (mm)	[HST_{de}]	400.0000	
Minimum wall thickness of honing wheel (mm)	[HST_{sR}]	12.5000	
Guide value for maximum honing wheel tip circle (mm)	[$HST_{da\ max}$]	-376.0000	
Maximum HST (honing stone) active root circle (mm)	[$HST_{dNf\ max}$]	-375.0000	
Proposed HST (honing stone) width (mm)	[HST_b]	44.0000	
Proposed oscillation path (mm)	[±HST_o]	0.5000	

14 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	153.123

14.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	0.830	2.598
Total mass (kg)	[Σm]	3.428	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.0005124	0.03229
System (da+df)/2...di (kg*m²)	[J]	0.002914	

14.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	0.598
Torsion when subjected to nominal torque (°)	[δcr]	0.072
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	84.450

14.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.065
Wear sliding coefficient by Niemann	[ζw]	1.067
Loss factor	[HV]	0.161
Gear power loss (kW)	[PVZ]	2.462
Meshing efficiency (%)	[ηz]	98.951

15 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	9500.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	7972

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	2.407e+04
Tooth flank service life (h)	[H_{Hatt}]	7972	2.923e+04

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

15.1 Damage

Damage relative to the required service life (H, 9500.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	39.4643	119.1658	32.4998

Damage relative to the system service life (H_{att}, 7972.1 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	33.1172	100.0000	27.2727

16 Reliability calculation

16.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

16.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	179016	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	179016	1.300e+00	1.286e+09	6.129e+09	95.9918
2	Tooth root	48823	1.700e+00	1.135e+09	1.744e+09	100.0000
2	Tooth flank	48823	1.300e+00	1.286e+09	6.129e+09	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

16.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	9500.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	95.9918
Reliability R, gears subsystem (%)	[R_{subG}]	95.9918
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	24072.3740
Service life H, tooth flanks subsystem (h)	[H_{subH}]	7972.0859
Service life H, gears subsystem (h)	[H_{subG}]	7972.0859

17 Remarks

17.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

17.2 Calculations and factors

- Calculation of Z_{β} acc. to Corrigendum 1 ISO 6336-2:2008 with $Z_{\beta} = 1/(COS(\beta)^{0.5})$
- Details of calculation method:
  c_{γ} according to Method B
  K_v according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 675)